

spurious correlations

correlation is not causation

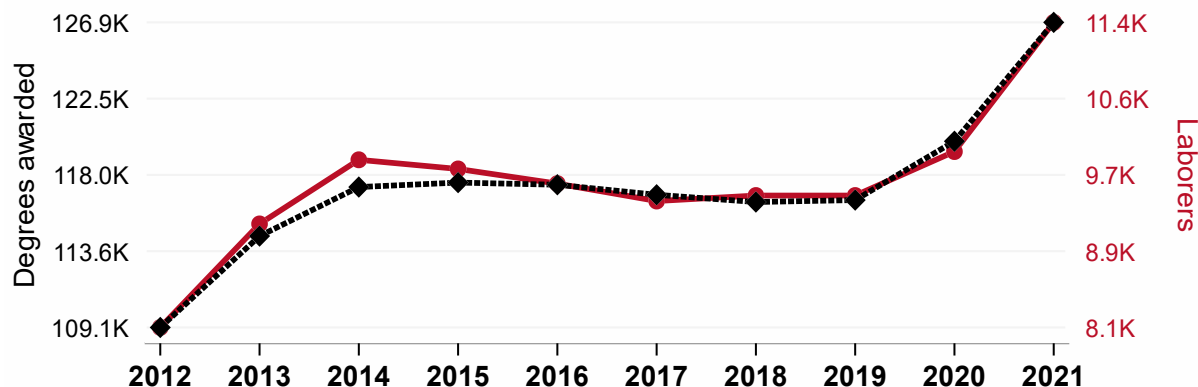
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don't miss [spurious scholar](#),
where each of these is an academic paper

Bachelor's degrees awarded in Psychology

correlates with

The number of groundskeepers in Utah



◆ Bachelor's degrees conferred by postsecondary institutions, in field of study: Psychology · Source: National Center for Education Statistics

● BLS estimate of landscaping and groundskeeping workers in Utah · Source: Bureau of Labor Statistics

2012-2021, $r=0.990$, $r^2=0.980$, $p<0.01$ · tylervigen.com/spurious/correlation/1781

[View details about correlation #1,781](#)

Minds Over Matter: The Psych-ing Connection Between Bachelor's Degrees in Psychology and Groundskeepers in Utah

Show GenAI's made-up explanation

As the number of psychology graduates rose, so did the demand for shrinks, er, I mean shrub experts. It seems like everyone was finally recognizing the deep-rooted issues in Utah's green spaces and decided to take a leaf out of Freud's book by tackling them head-on. As these graduates mulled over their career options, they realized that they could cultivate a fulfilling career in groundskeeping, digging deep into the soil of the human psyche while also mowing lawns. It's a Freudian field of work indeed, where they can help grass cope with its existential crises and provide much-needed therapy for plants experiencing growth issues. It's safe to say that these psychology grads are really raking in the success in the world of

horticulture, proving that when it comes to tending to the state of Utah's landscapes, they're not just whistling dixie!

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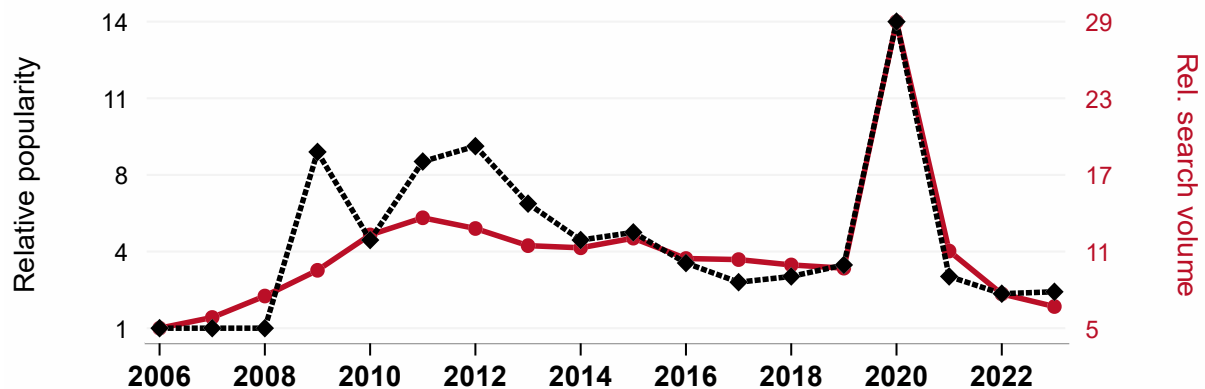
What else correlates?

Bachelor's degrees awarded in Psychology · all education
The number of groundskeepers in Utah · all occupations

Popularity of the 'like a boss' meme

correlates with

Google searches for 'how to cut own hair'



◆--- Relative volume of Google searches for 'like a boss' (without quotes, in the United States) · Source: Google Trends

—●— Relative volume of Google searches for 'how to cut own hair' (Worldwide, without quotes) · Source: Google Trends

2006-2023, $r=0.871$, $r^2=0.758$, $p<0.01$ · [tylervigen.com/spurious/correlation/4994](https://www.tylervigen.com/spurious/correlation/4994)

[View details about correlation #4,994](#)

Hair Today, Like a Boss Tomorrow: The Curious Connection Between 'Like a Boss' Meme Popularity and DIY Haircut Searches

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[Show scatterplot](#)

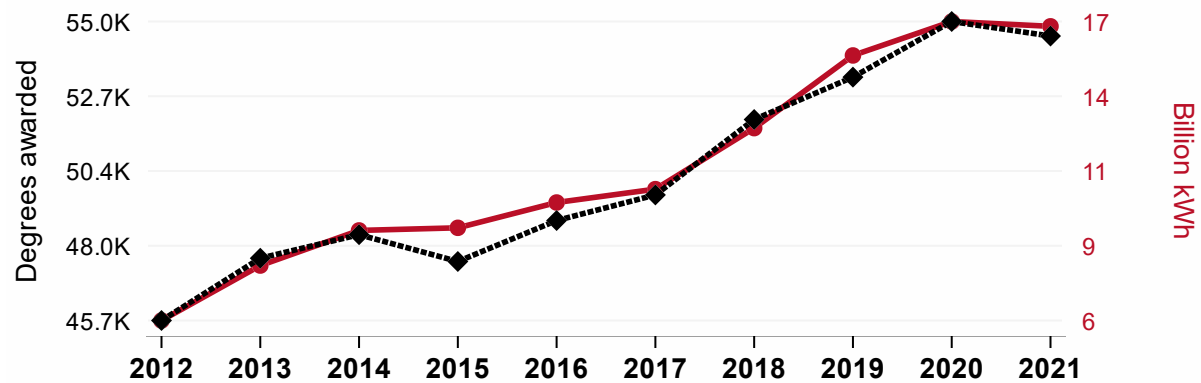
What else correlates?

Popularity of the 'like a boss' meme · all memes
Google searches for 'how to cut own hair' · all google searches

Bachelor's degrees awarded in interdisciplinary studies

correlates with

Electricity generation in Angola



◆ Bachelor's degrees conferred by postsecondary institutions, in field of study: Multi/interdisciplinary studies · Source: National Center for Education Statistics

● Total electricity generation in Angola in billion kWh · Source: Energy Information Administration

2012-2021, $r=0.992$, $r^2=0.984$, $p<0.01$ · [tylervigen.com/spurious/correlation/2666](https://www.tylervigen.com/spurious/correlation/2666)

View details about correlation #2,666

Voltage Variety: The Shocking Connection Between Multi/Interdisciplinary Studies Bachelor's Degrees and Electricity Generation in Angola

[Show GenAI's made-up explanation](#)

As more students delved into the vast and varied world of Multi/interdisciplinary studies, they sparked a surge in brainpower. This intellectual firestorm led to an influx of innovative ideas for improving efficiency in electricity generation. It's shocking how these interdisciplinary graduates connected the dots to current flow in Angola's power sector, igniting a revolution in energy production. You could say they truly sparked a bright idea or two!

Show GenAI image*Show scatterplot**What else correlates?*

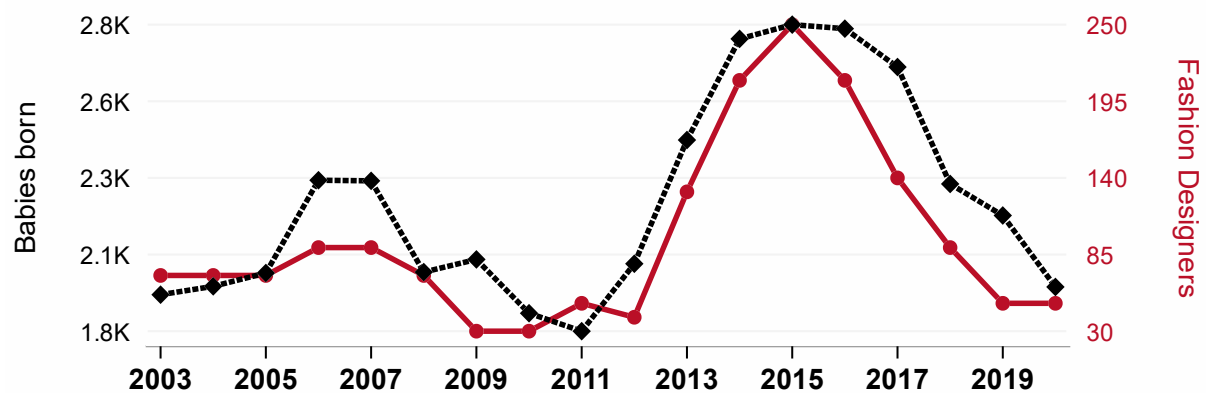
Bachelor's degrees awarded in interdisciplinary studies · all education

Electricity generation in Angola · all energy

Popularity of the first name Ariel

correlates with

The number of fashion designers in Michigan



◆--- Babies of all sexes born in the US named Ariel · Source: US Social Security Administration

●— BLS estimate of fashion designers in Michigan · Source: Bureau of Labor Statistics

2003-2020, $r=0.924$, $r^2=0.854$, $p<0.01$ · [tylervigen.com/spurious/correlation/2985](https://www.tylervigen.com/spurious/correlation/2985)

View details about correlation #2,985

Ariels in Michigan: Making Waves in Fashion Design

Show GenAI's made-up explanation

[Show GenAI image](#)[Show scatterplot](#)*What else correlates?*

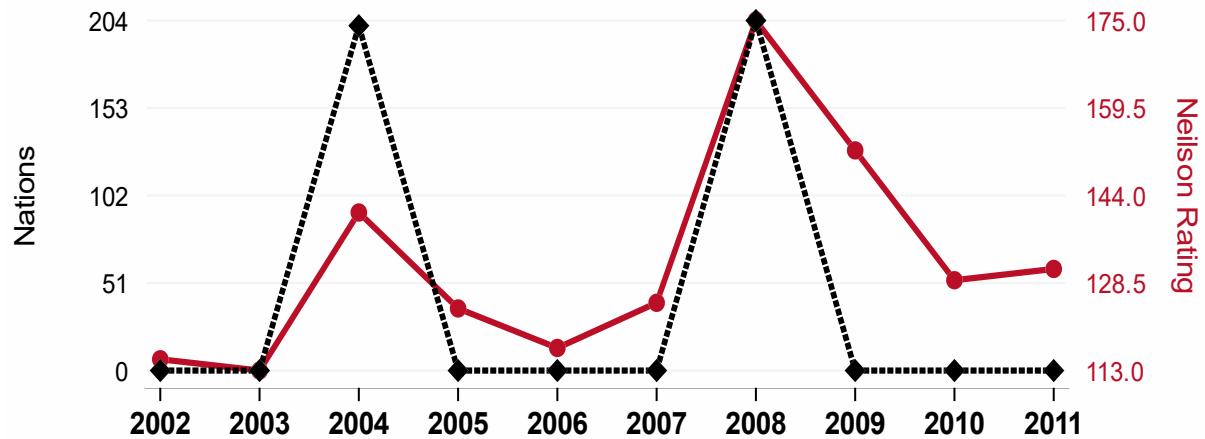
Popularity of the first name Ariel · all first names

The number of fashion designers in Michigan · all occupations

Number of competing nations in the Summer Olympics

correlates with

Nielsen Ranking of Smallville Season Finale



◆--- Number of competing nations in the Summer Olympics · Source: Wikipedia

●— Nielsen Ranking of Smallville Season Finales by Air Year · Source: Wikipedia

2002-2011, $r=0.711$, $r^2=0.505$, $p<0.05$ · tylervigen.com/spurious/correlation/2648**[View details about correlation #2,648](#)***The Olympics Effect: The Interplay between International Sporting Events and Small-Town Dramas*[Show GenAI's made-up explanation](#)[Show GenAI image](#)[Show scatterplot](#)*What else correlates?*

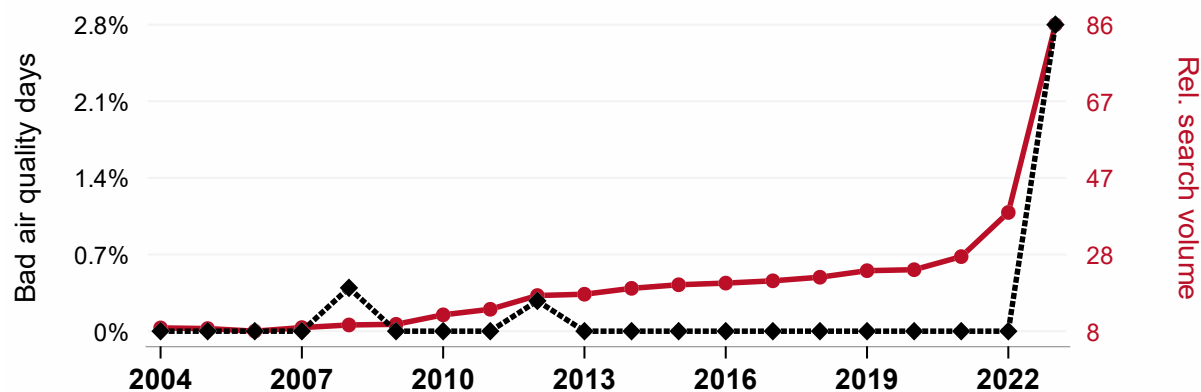
Number of competing nations in the Summer Olympics · all weird & wacky

Nielsen Ranking of Smallville Season Finale · all weird & wacky

Air pollution in Appleton, Wisconsin

correlates with

Google searches for 'ice bath'



◆ Percentage of days 'unhealthy' or worse air quality in Appleton, WI

· Source: Environmental Protection Agency

● Relative volume of Google searches for 'ice bath' (Worldwide, without quotes) · Source: Google Trends

2004-2023, $r=0.867$, $r^2=0.752$, $p<0.01$ · tylervigen.com/spurious/correlation/2867

View details about correlation #2,867

Chilling Consequences: Uncovering the Icy Relationship between Air Pollution and 'Ice Bath' Google Searches in Appleton, Wisconsin

Show GenAI's made-up explanation

As the air quality dipped, residents of Appleton, Wisconsin found themselves taking 'chill pills' in the form of ice baths. It seems like the smog just couldn't take a hint and decided to 'mist' with people's routines, leading them to seek refreshing relief in a tub of ice. It's a classic case of pollution causing a cool-ution, as the city's air became more inhospitable, driving folks to 'cool off' in the most literal way possible. Who knew that as the pollution levels rose, so did the temptation to take the plunge into an icy situation? It's enough to make you say, "What in the world is 'smog-nifying' this frosty trend?" But hey, when the going gets 'foggy,' the foggy get going... straight to the freezer for an impromptu arctic dip.

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Show scatterplot

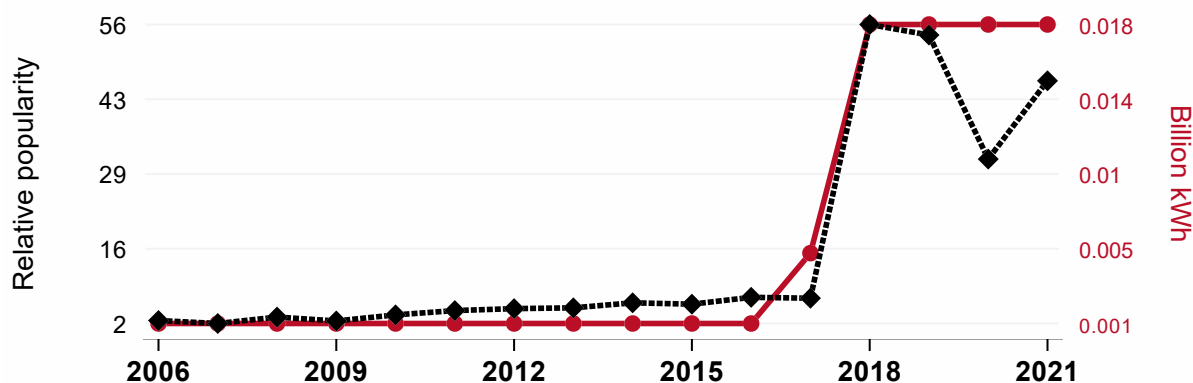
What else correlates?

Air pollution in Appleton, Wisconsin · all weather
Google searches for 'ice bath' · all google searches

Popularity of the 'we live in a society' meme

correlates with

Wind power generated in Namibia



◆ Relative volume of Google searches for 'we live in a society' (without quotes, in the United States) · Source: Google Trends

● Total wind power generated in Namibia in billion kWh · Source: Energy Information Administration

2006-2021, $r=0.961$, $r^2=0.924$, $p<0.01$ · [tylervigen.com/spurious/correlation/4950](https://www.tylervigen.com/spurious/correlation/4950)

View details about correlation #4,950

We Live in a Current-Generating Society: The Shocking Connection Between 'We Live in a Society' Meme Popularity and Wind Power Generated in Namibia

[Show GenAI's made-up explanation](#)[Show GenAI image](#)[Show scatterplot](#)

What else correlates?

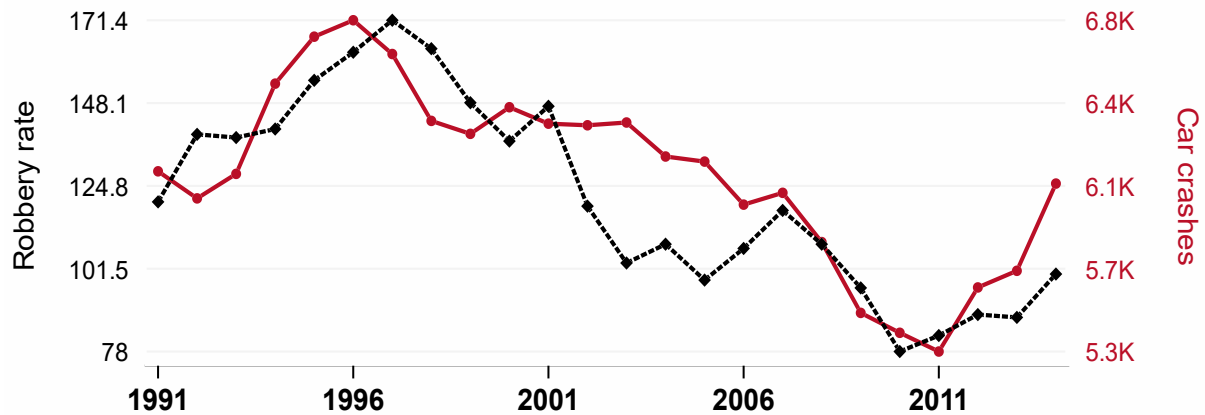
Popularity of the 'we live in a society' meme · all memes

Wind power generated in Namibia · all energy

Robberies in New Mexico

correlates with

Car crashes in the US



◆--- The robbery rate per 100,000 residents in New Mexico · Source: FBI Criminal Justice Information Services

—● Total annual vehicle crash count in the US · Source: Statista

1991-2014, $r=0.835$, $r^2=0.697$, $p<0.01$ · tylervigen.com/spurious/correlation/3981

[View details about correlation #3,981](#)

The Highwaymen: Exploring the Hold-Up between Robberies in New Mexico and Car Crashes in the US

[Show GenAI's made-up explanation](#)[Show GenAI image](#)[Show scatterplot](#)

What else correlates?

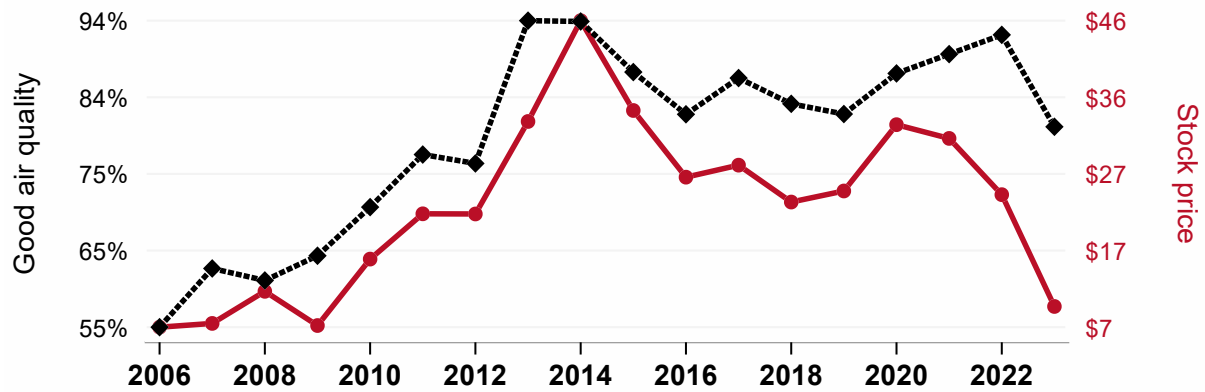
Robberies in New Mexico · all random state specific

Car crashes in the US · all weird & wacky

Air quality in Vineland, New Jersey

correlates with

Warner Bros. Discovery's stock price (WBD)



◆-- Percentage of days with good air quality in Vineland-Bridgeton, NJ · Source: Environmental Protection Agency

●— Opening price of Warner Bros. Discovery (WBD) on the first trading day of the year · Source: LSEG Analytics (Refinitiv)

2006-2023, $r=0.867$, $r^2=0.751$, $p<0.01$ · tylervigen.com/spurious/correlation/2929

View details about correlation #2,929

A Breath of Fresh Air: Examining the Atmospheric Impact on Warner Bros. Discovery's Stock Price in Vineland, New Jersey

Show GenAI's made-up explanation

Show GenAI image

Show scatterplot

What else correlates?

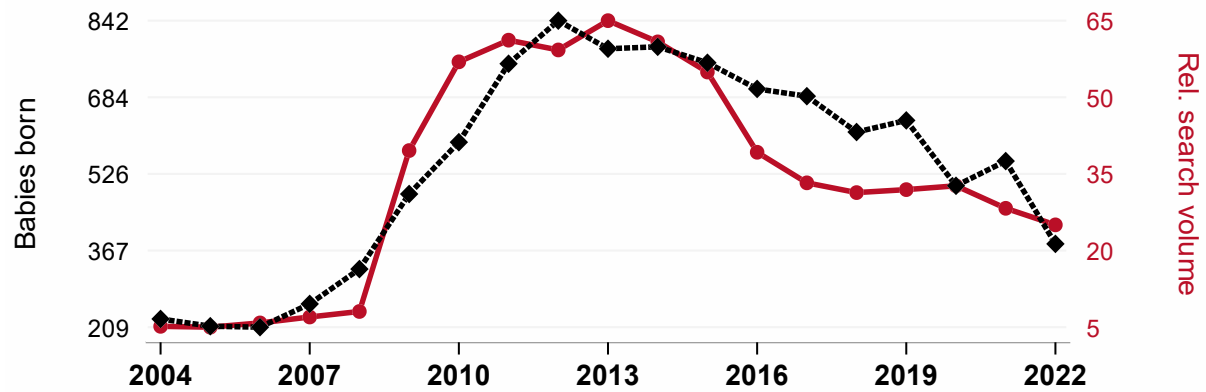
Air quality in Vineland, New Jersey · all weather

Warner Bros. Discovery's stock price (WBD) · all stocks

Popularity of the first name Dexter

correlates with

Google searches for 'bing'



◆--- Babies of all sexes born in the US named Dexter · Source: US Social Security Administration

●— Relative volume of Google searches for 'bing' (Worldwide, without quotes) · Source: Google Trends

2004-2022, $r=0.921$, $r^2=0.847$, $p<0.01$ · tylervigen.com/spurious/correlation/5230

View details about correlation #5,230

Dexter's Dichotomy: Delving into the Data Doldrums of 'Bing' and Baby Names

[Show GenAI's made-up explanation](#)

[Show GenAI image](#)

[Show scatterplot](#)

What else correlates?

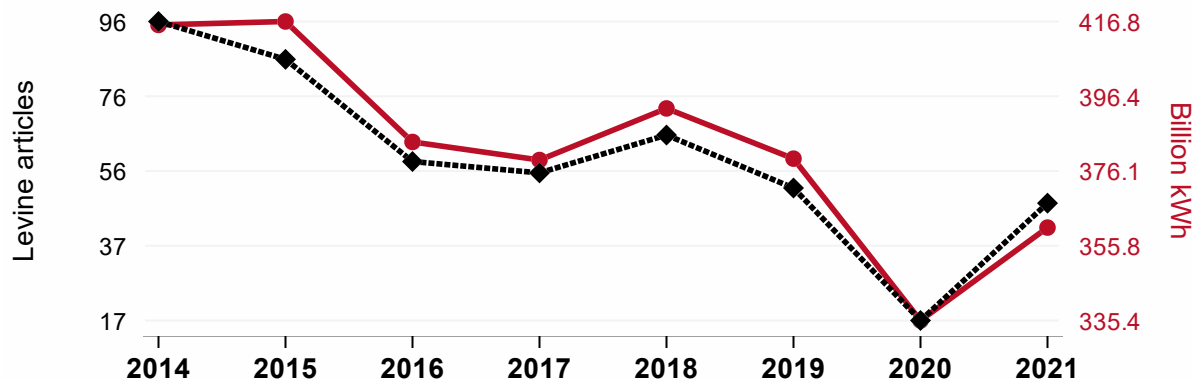
Popularity of the first name Dexter · all first names

Google searches for 'bing' · all google searches

Number of articles Matt Levine published on Bloomberg on Wednesdays

correlates with

Nuclear power generation in France



◆ Articles published by Matt Levine on Bloomberg.com on Wednesday · Source: Bloomberg

● Total nuclear power generation in France in billion kWh · Source: Energy Information Administration

2014-2021, $r=0.980$, $r^2=0.961$, $p<0.01$ · tylervigen.com/spurious/correlation/5894

View details about correlation #5,894

Watt's Up, Matt? The Electrifying Connection Between Matt Levine's Wednesday Bloomberg Articles and Nuclear Power Generation in France

Show GenAI's made-up explanation

Apparently, Matt's puns were so powerful, they were causing a chain reaction in the French energy sector. Who knew finance and fission had such a strong connection? Keep an eye out for his next article - it might just be the catalyst for a meltdown!

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[Show scatterplot](#)*What else correlates?*

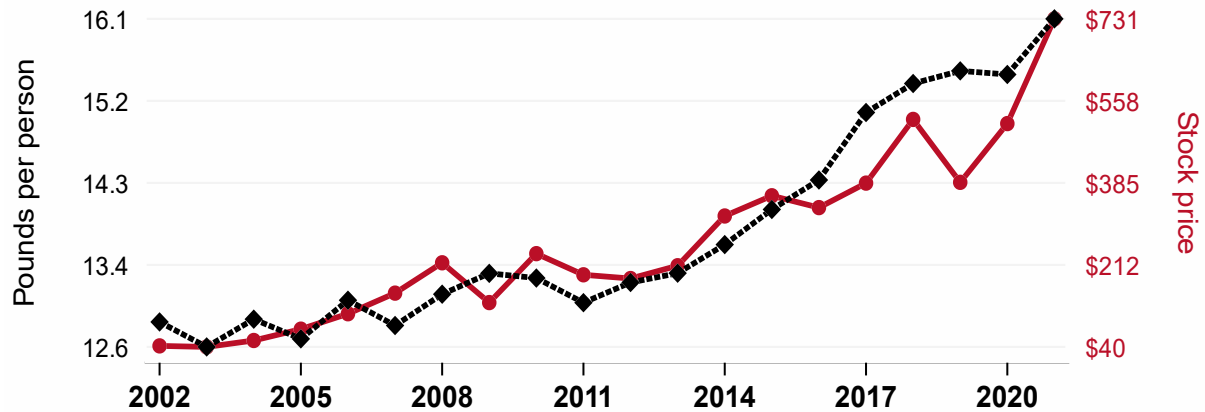
Number of articles Matt Levine published on Bloomberg on Wednesdays · all weird & wacky

Nuclear power generation in France · all energy

American cheese consumption

correlates with

BlackRock's stock price (BLK)



◆ Per capita consumption of American cheese in the US · Source: USDA

● Opening price of BlackRock (BLK) on the first trading day of the year · Source: LSEG Analytics (Refinitiv)

2002-2021, $r=0.940$, $r^2=0.883$, $p<0.01$ · tylervigen.com/spurious/correlation/4018**[View details about correlation #4,018](#)***Cheddar and Cheddar: The Cheesy Connection Between American Cheese Consumption and BlackRock's Stock Price*[Show GenAI's made-up explanation](#)[Show GenAI image](#)[Show scatterplot](#)*What else correlates?*

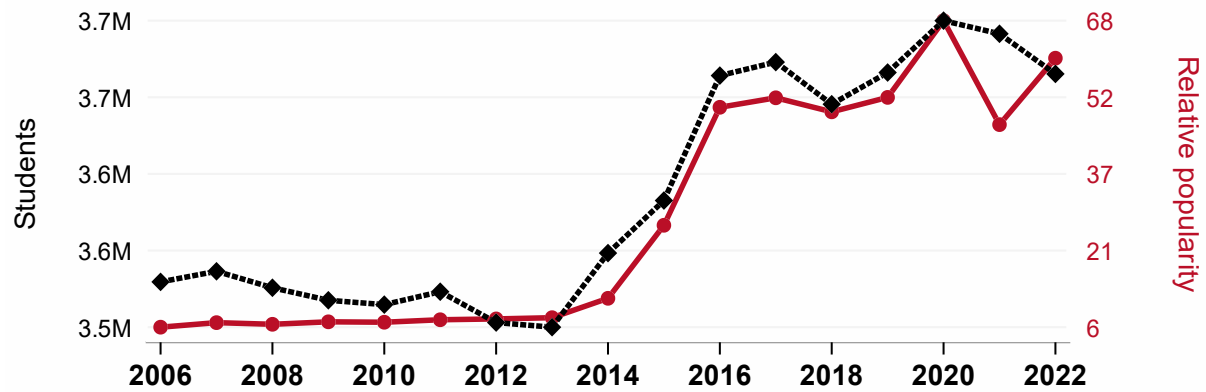
American cheese consumption · all food

BlackRock's stock price (BLK) · all stocks

Number of public school students in 11th grade

correlates with

Popularity of the 'this is fine' meme



◆ 11th grade enrollment in public elementary and secondary schools · Source: National Center for Education Statistics

● Relative volume of Google searches for 'this is fine' (without quotes, in the United States) · Source: Google Trends

2006-2022, $r=0.975$, $r^2=0.951$, $p<0.01$ · tylervigen.com/spurious/correlation/5953

View details about correlation #5,953

The Prevalence of Poignant Proportions: Public School Pupils in 11th grade and the Popularity of the 'This is Fine' Meme

Show GenAI's made-up explanation

Show GenAI image

Show scatterplot

What else correlates?

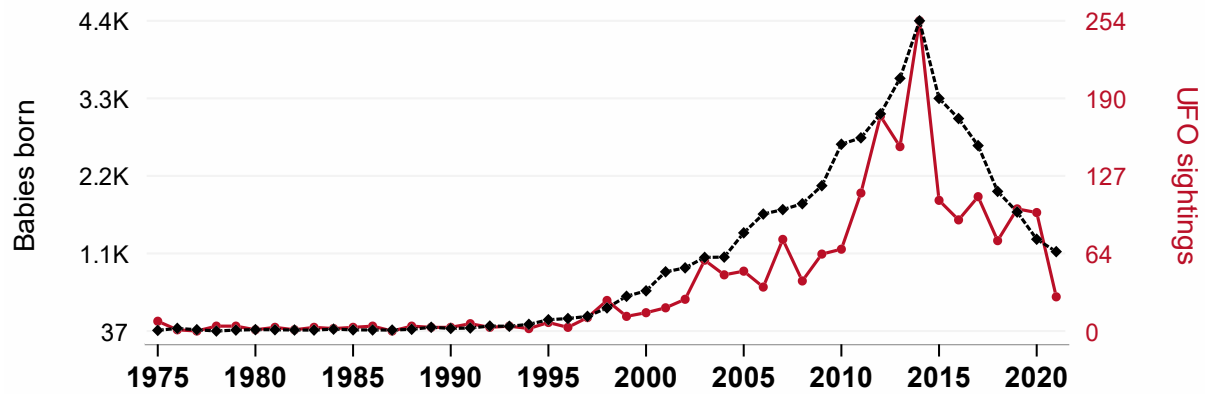
Number of public school students in 11th grade · all education

Popularity of the 'this is fine' meme · all memes

Popularity of the first name Annabelle

correlates with

UFO sightings in South Carolina



◆--- Babies of all sexes born in the US named Annabelle · Source: US Social Security Administration

—●— UFO sightings reported in South Carolina · Source: National UFO Reporting Center

1975-2021, $r=0.931$, $r^2=0.867$, $p<0.01$ · tylervigen.com/spurious/correlation/2085

View details about correlation #2,085

Unidentified First Name Phenomena: The Anns and Aliens Connection in South Carolina

Show GenAI's made-up explanation

It's simple, really. As more parents named their daughters Annabelle after the creepy doll in the horror movies, they inadvertently unleashed a wave of spooky energy. This supernatural surge somehow attracted UFOs to South Carolina, creating a breeding ground for otherworldly encounters. Remember, when it rains popularity, it pours unidentified flying objects!

Show GenAI image



Show scatterplot

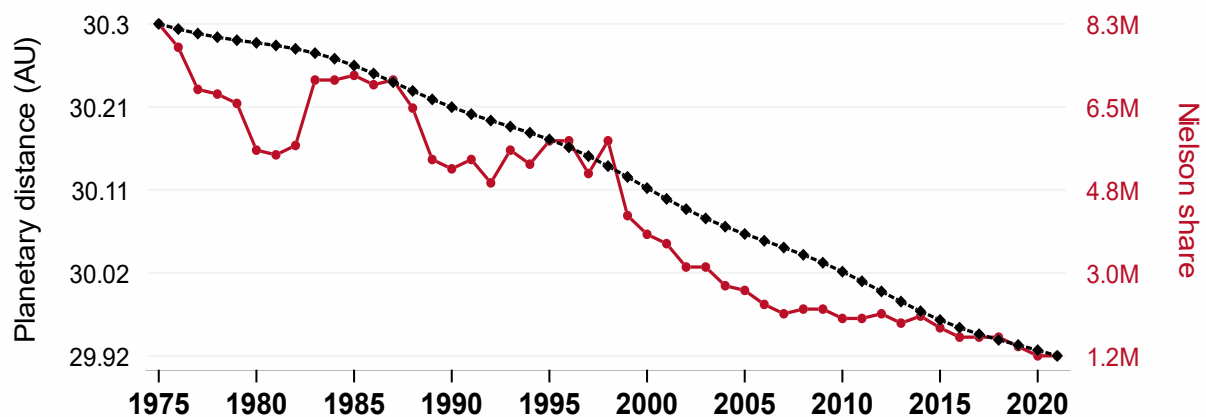
What else correlates?

Popularity of the first name Annabelle · all first names
UFO sightings in South Carolina · all random state specific

The distance between Neptune and the Sun

correlates with

Viewership count for Days of Our Lives



◆--- The average distance between Neptune and the Sun as measured on the first day of each month · Source: Calculated using Astropy

—● Metered viewership count for Days of Our Lives · Source: Wikipedia

1975-2021, $r=0.959$, $r^2=0.920$, $p<0.01$ · tylervigen.com/spurious/correlation/1034

View details about correlation #1,034

Lost in Space Opera: An Interstellar Analysis of Neptune's Distance from the Sun and its Impact on Days of Our Lives Viewership

[Show GenAI's made-up explanation](#)[Show GenAI image](#)[Show scatterplot](#)

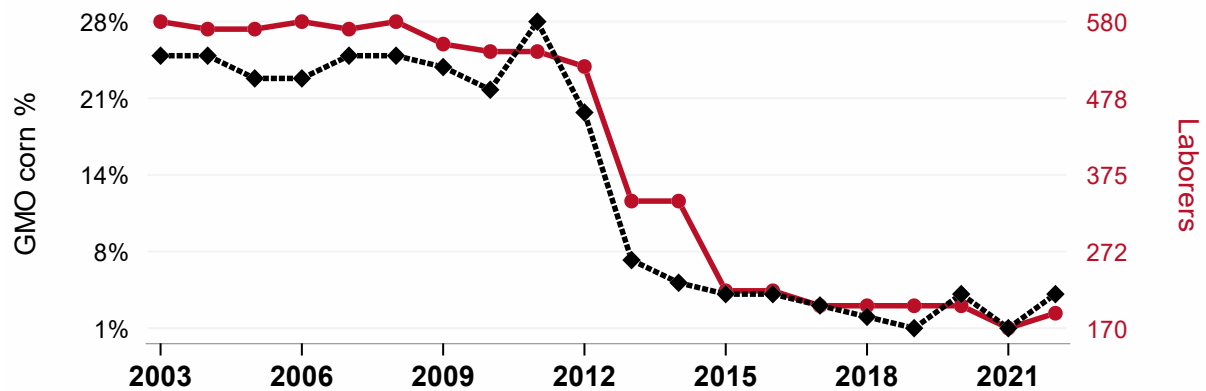
What else correlates?

The distance between Neptune and the Sun · all planets
 Viewership count for Days of Our Lives · all weird & wacky

GMO use in corn grown in Kansas

correlates with

The number of postmasters in Kansas



◆--- Percent of all corn planted in Kansas that is genetically modified to be insect-resistant (Bt), but not herbicide-tolerant (HT) · Source: USDA

●— BLS estimate of postmasters and mail superintendents in Kansas · Source: Bureau of Labor Statistics

2003-2022, $r=0.978$, $r^2=0.956$, $p<0.01$ · [tylervigen.com/spurious/correlation/1744](https://www.tylervigen.com/spurious/correlation/1744)

View details about correlation #1,744

The Corny Connection: The Correlation Between GMO Corn and Postmaster Proliferation in Kansas

[Show GenAI's made-up explanation](#)[Show GenAI image](#)[Show scatterplot](#)

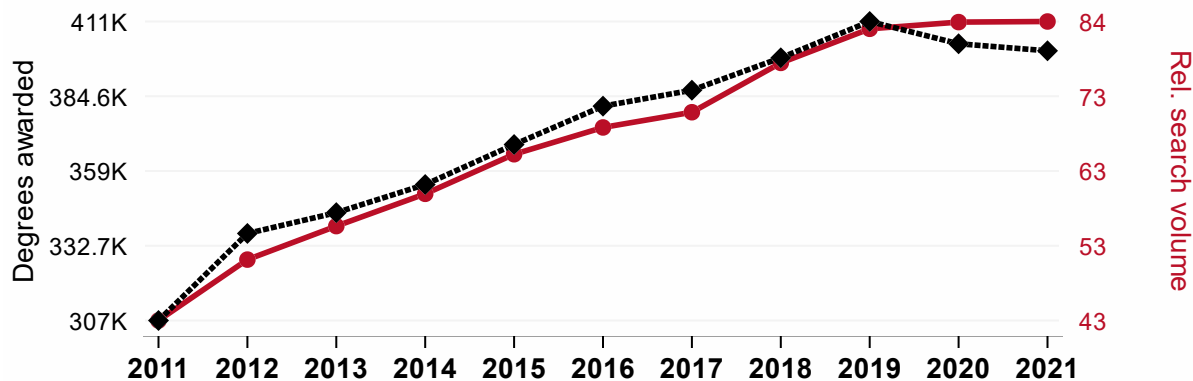
What else correlates?

GMO use in corn grown in Kansas · all food
 The number of postmasters in Kansas · all occupations

Associates degrees awarded in Liberal arts

correlates with

Google searches for 'tummy ache'



◆ Associate's degrees conferred by postsecondary institutions with a field of study of Liberal arts and sciences, general studies, and humanities · Source: National Center for Education Statistics

● Relative volume of Google searches for 'tummy ache' (Worldwide, without quotes) · Source: Google Trends

2011-2021, $r=0.987$, $r^2=0.975$, $p<0.01$ · tylervigen.com/spurious/correlation/1530

View details about correlation #1,530

The Liberal Arts Laughs: A Gut-Busting Investigation into the Correlation between Associates Degrees in Liberal Arts and Google Searches for 'Tummy Ache'

Show GenAI's made-up explanation

As more people delved into the depths of philosophy and existential thinking, the collective realization of the absurdity of human existence led to a wave of stomach-churning existential dread and questioning of the universe's grand design. This, in turn, manifested as a notable uptick in tummy ache-related Google searches as individuals grappled with the cosmic uncertainty while pondering the true nature of being. It seems that pondering the meaning of life may have led to some uneasy feelings in the pit of many a contemplative stomach.

Show GenAI image



Show scatterplot

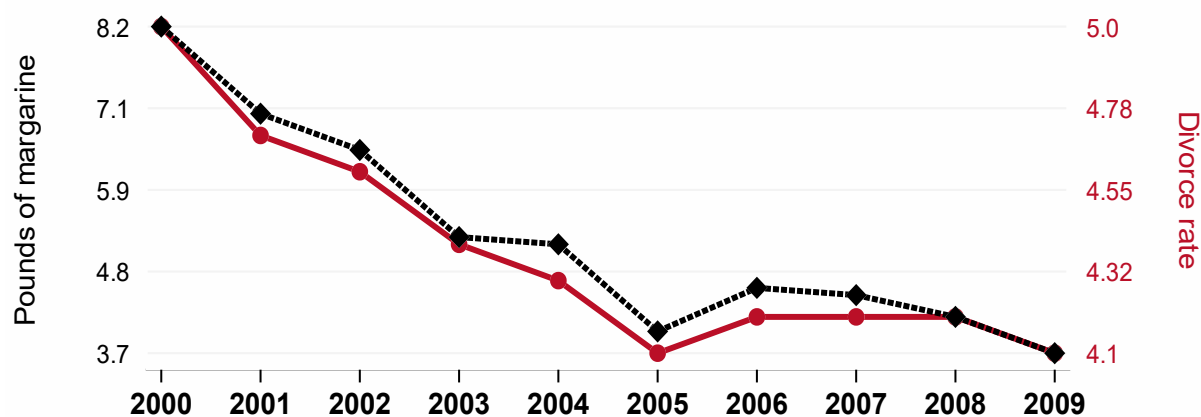
What else correlates?

Associates degrees awarded in Liberal arts · all education
Google searches for 'tummy ache' · all google searches

Per capita consumption of margarine

correlates with

The divorce rate in Maine



◆ Per capita consumption of margarine in the United States · Source: US Department of Agriculture

● The divorce rate in Maine · Source: CDC National Vital Statistics

2000-2009, $r=0.993$, $r^2=0.985$, $p<0.01$ · tylervigen.com/spurious/correlation/5920

View details about correlation #5,920

Spreading Love and Margarine: An Examination of the Butter-Splitter Correlation in Maine

[Show GenAI's made-up explanation](#)[Show GenAI image](#)[Show scatterplot](#)

What else correlates?

Per capita consumption of margarine · all food
The divorce rate in Maine · all random state specific

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Why this works

1. **Data dredging:** I have 25,237 variables in my database. I compare all these variables against each other to find ones that randomly match up. That's 636,906,169 correlation calculations! This is called "**data dredging**".^{Note} Instead of starting with a hypothesis and testing it, I instead tossed a bunch of data in a blender to see what correlations would shake out. It's a dangerous way to go about analysis, because any sufficiently large dataset will yield strong correlations completely at random.
2. **Lack of causal connection:** There is probably no direct connection between these variables, despite what the AI says above.^{Note} This is exacerbated by the fact that I used "Years" as the base variable. Lots of things happen in a year that are not related to each other! Most studies would use something like "one person" in stead of "one year" to be the "thing" studied.
3. **Observations not independent:** For many variables, sequential years are not independent of each other. You will often see trend-lines form. If a population of people is continuously doing something every day, there is no reason to think they would suddenly *change* how they are doing that thing on January 1. A naive *p*-value calculation does not take this into account.^{Note}
4. **Y-axes doesn't start at zero:** I truncated the Y-axes of the graphs above. I also used a line graph, which makes the visual connection stand out more than it deserves.^{Note} Mathematically what I showed is true, but it is intentionally misleading. If you click on any of the charts that abuse this, you can scroll down to see a version that starts at zero.

5. **Confounding variable:** Confounding variables (like global pandemics) will cause two variables to look connected when in fact a "sneaky third" variable is influencing both of them behind the scenes.
6. **Outliers:** Some datasets here have outliers which drag up the correlation.^{Note} I intentionally mishandled outliers, which makes the correlation look extra strong.
7. **Low *n*:** There are not many data points included in some of these charts.^{Note} Even if the p-value is high, we should be suspicious of using so few datapoints in a correlation.

Pro-tip: click on any correlation to see:

- Detailed data sources
- Prompts for the AI-generated content
- Explanations of each of the calculations (correlation, p-value)
- Python code to calculate it yourself



Project by Tyler Vigen

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